Lec-15

11/04/26 12:54 PM

Func calling itself.

A() {
A() ⇒
}

fun calling itself

Funcalling B() { }

Funcalling A() { B() }

PMI

$$\sum_{i=1}^{i=N} i = 1+2+3+4+\dots N = \frac{N(N+1)}{2}$$

L.H.S R.H.S

$$\sum_{i=1}^{i=1} i = 1 \Rightarrow \frac{1\times(1+1)}{2}$$

$$\sum_{i=1}^{i=2} i = 1+2 \Rightarrow \frac{2\times(2+1)}{2}$$

$$\sum_{i=1}^{i=K} i = 1+2+3+\dots+K = \frac{K(K+1)}{2}$$ ① ✓

Recursion

① Smaller input ✓
⑪ K → correct ✓
⑪ K+1 → prove ✓

$$\sum_{i=1}^{i=K+1} i = 1+2+3+\dots K+K+1 = \frac{(K+1)(K+2)}{2}$$

$$\frac{K\times(K+1)}{2} + (K+1) =$$

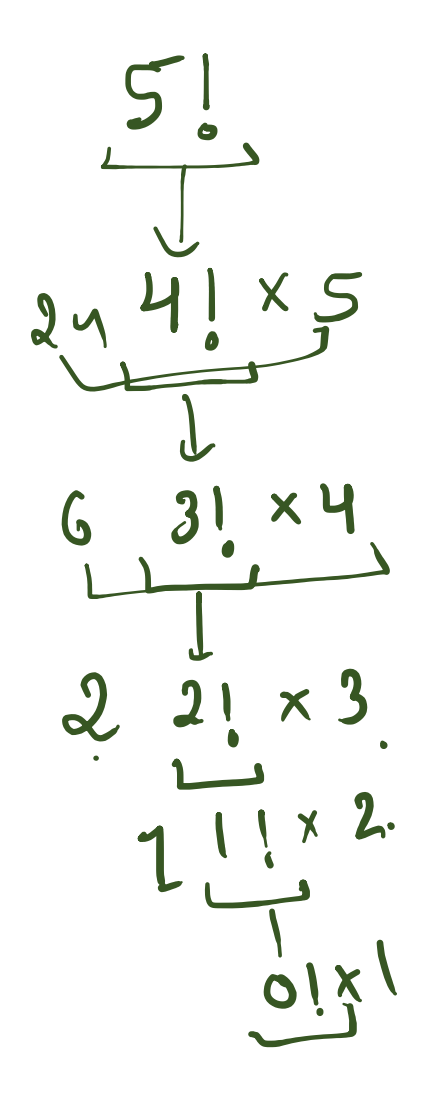
$$(K+1)\left[\frac{K}{2}+1\right] =$$

$$(K+1)\frac{(K+2)}{2}$$

A() {
→ A()
}

Smaller problem
Base case

1.
2.
3. 2nd step + self work
B.P

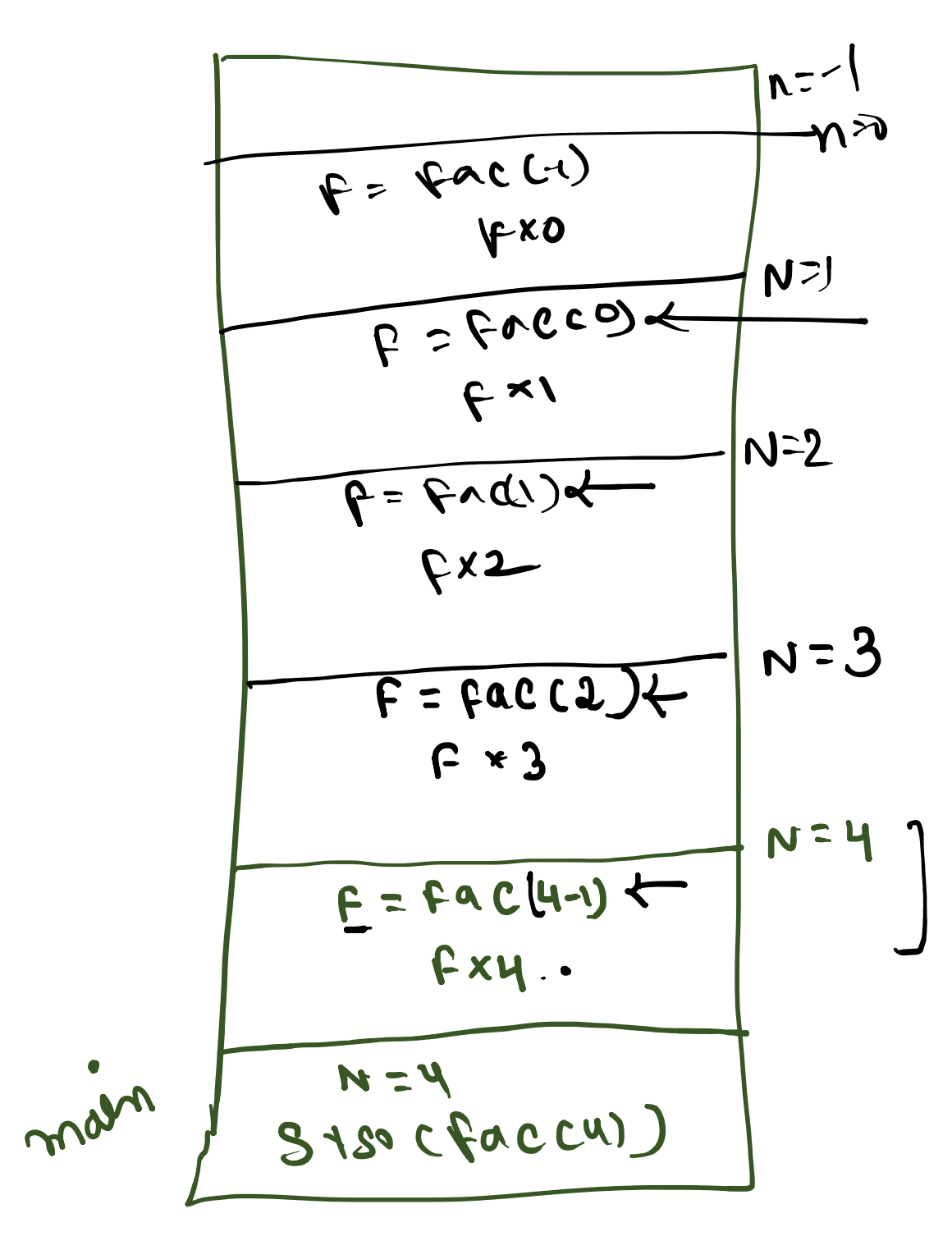

5!
24 4! × 5
6 3! × 4
2 2! × 3
1 1! × 2
0! × 1

$N! \to (N-1)! \times N$

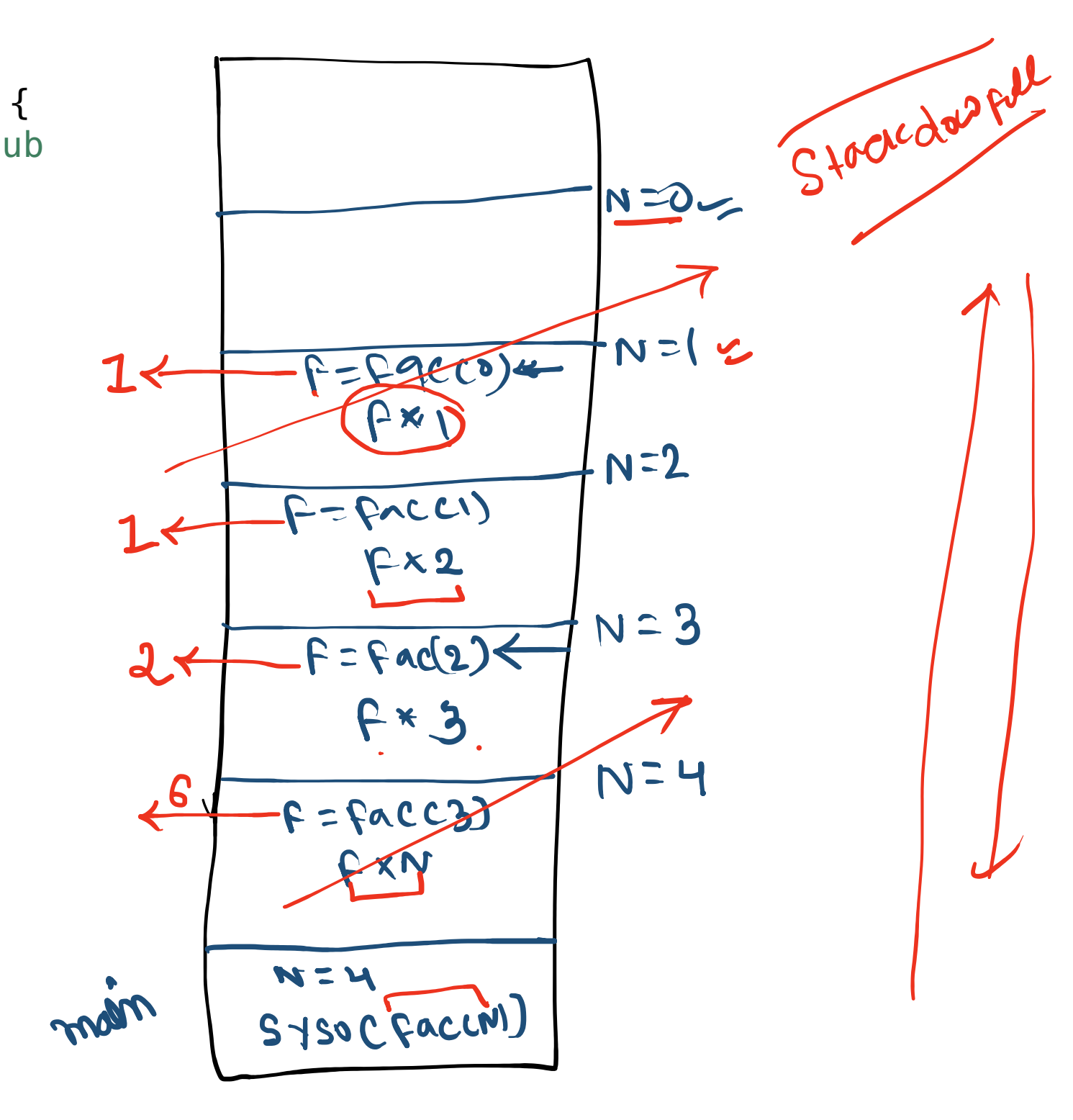
$n \to (n-1) + n$

$a^n \to (a^{n-1}) \times a$

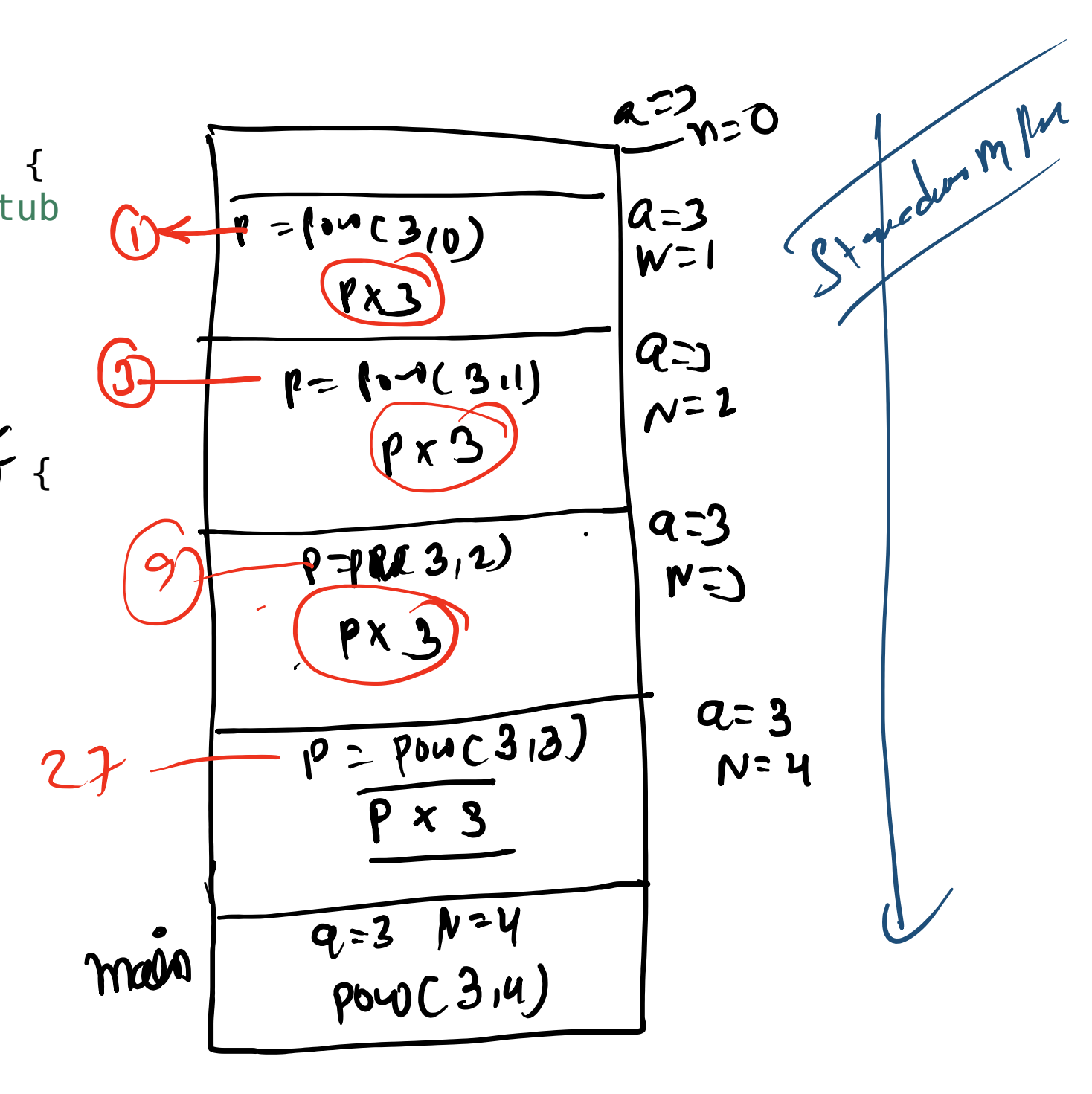
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 4;
        System.out.println(fac(n));

    }

    public static int fac(int n) {

        int f = fac(n - 1);// SP
        return f * n;
    }
```


n=-1
n=0
f = fac(-1)
f×0
N=1
f = fac(0)
f×1
N=2
f = fac(1)
f×2
N=3
f = fac(2)
f * 3
N=4
f = fac(4-1)
f×4
main
N=4
Syso(fac(4))

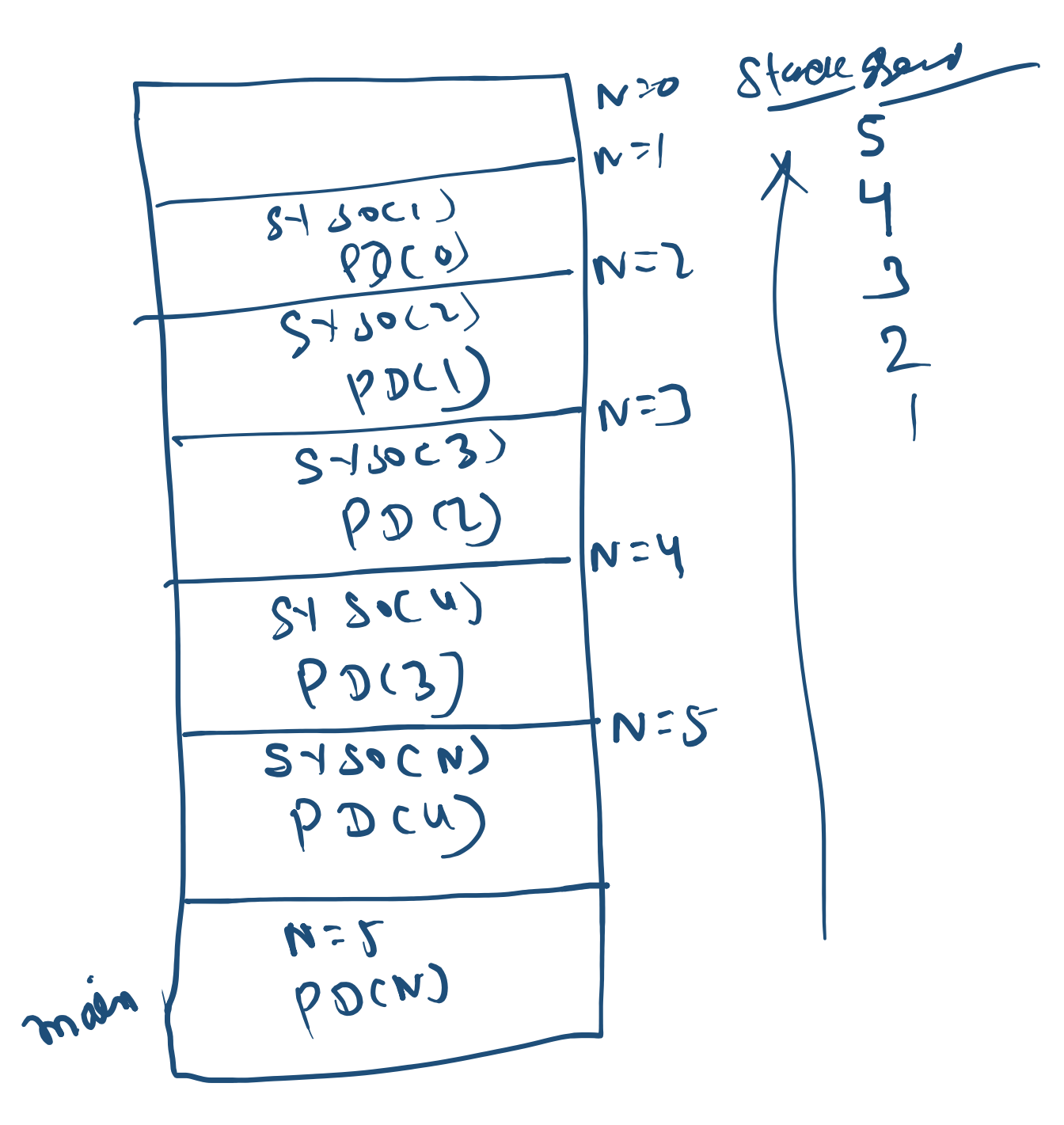
```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 4;
        System.out.println(fac(n));

    }

    public static int fac(int n) {
        //base case
        if(n==0) {
            return 1;
        }

        int f = fac(n - 1);// SP
        return f * n;

    }
```


Stackdownpull
N=0
1
f = fac(0)
f×1
N=1
1
f = fac(1)
f×2
N=2
2
f = fac(2)
f × 3
N=3
6
f = fac(3)
f×N
N=4
main
N=4
Syso(fac(N))

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int a = 3;
        int n = 4;
        System.out.println(pow(a, n));

    }

    public static int pow(int a, int n) {
        if (n == 0) {
            return 1;
        }
        int p = pow(a, n - 1);
        return p * a;

    }
```

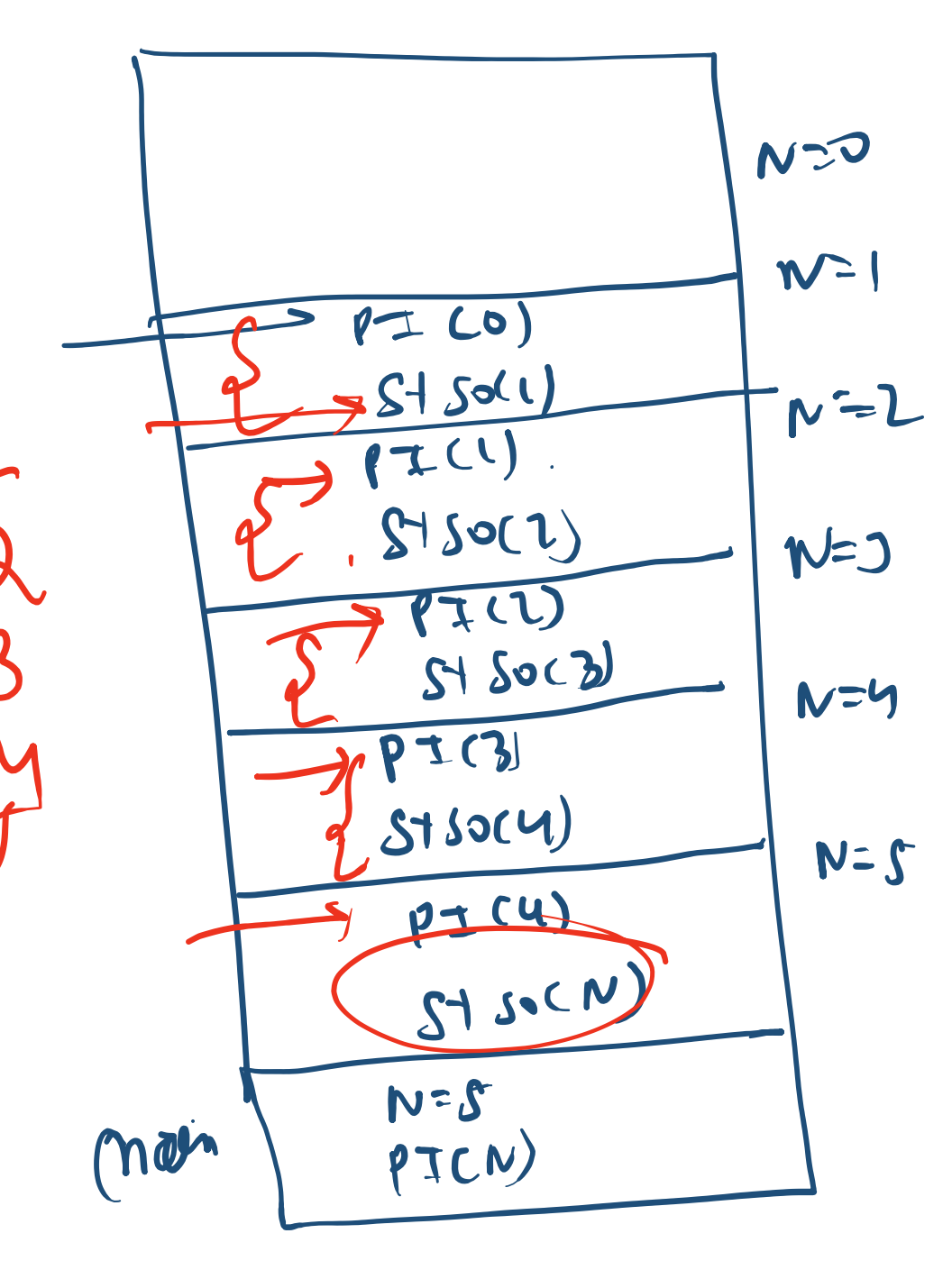

a=3
n=0
1
p = pow(3,0)
p×3
a=3
N=1
3
p = pow(3,1)
p×3
a=3
N=2
9
p = pow(3,2)
p×3
a=3
N=3
27
p = pow(3,3)
p × 3
a=3
N=4
main
a=3 N=4
pow(3,4)
Stack down m pull

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        PD(n);

    }

    public static void PD(int n) {
        // TODO Auto-generated method stub
        if (n == 0) {
            return;
        }
        System.out.println(n);
        PD(n - 1);

    }
```

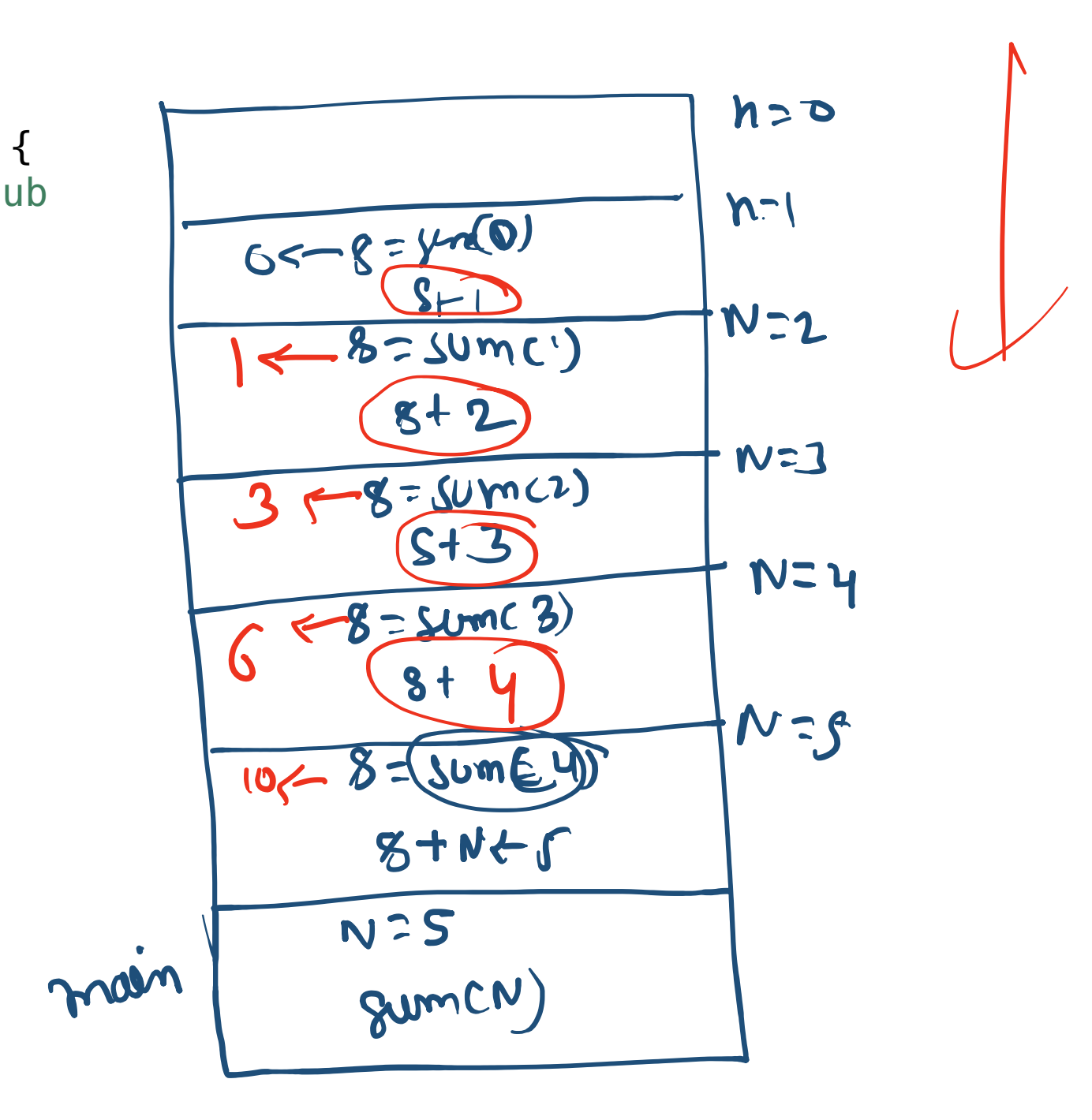

N=0
N=1
Stack Band
5
4
3
2
1
Syso(1)
PD(0)
N=2
Syso(2)
PD(1)
N=3
Syso(3)
PD(2)
N=4
Syso(4)
PD(3)
N=5
Syso(N)
PD(4)
main
N=5
PD(N)

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        PI(n);

    }

    public static void PI(int n) {
        if (n == 0) {
            return;
        }
        PI(n-1);
        System.out.println(n);

    }
```

1
2
3
4

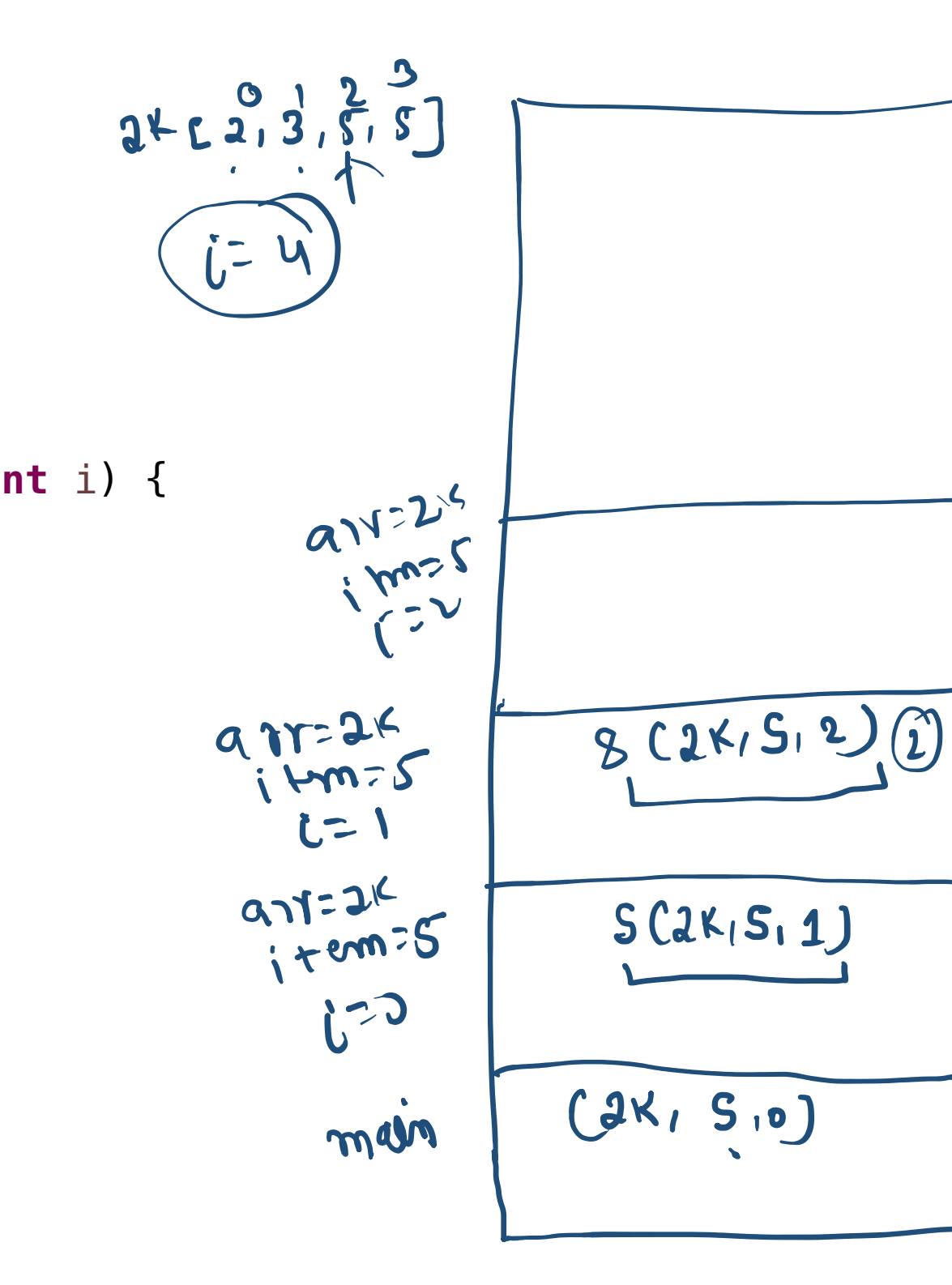

N=0
N=1
PI(0)
Syso(1)
N=2
PI(1)
Syso(2)
N=3
PI(2)
Syso(3)
N=4
PI(3)
Syso(4)
N=5
PI(4)
Syso(N)
main
N=5
PI(N)

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int n = 5;
        System.out.println(sum(n));

    }

    public static int sum(int n) {
        if(n==0) {
            return 0;
        }

        int s=sum(n-1);
        return s+n;

    }
```


n=0
n=1
0
s = sum(0)
s+1
N=2
1
s = sum(1)
s+2
N=3
3
s = sum(2)
s+3
N=4
6
s = sum(3)
s+4
N=5
10
s = sum(4)
s+N+5
main
N=5
sum(N)

i=0
0 1 2 3 4
[2, 3, 5, 4, 5] item: 5
i i

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int[] arr = { 2, 3, 5, 4, 5 };
        int item = 5;
        System.out.println(search(arr, item, 0));

    }

    public static int search(int[] arr, int item, int i) {

        if(arr[i]==item) {
            return i;
        }
        return search(arr, item, i+1);

    }
```


2k [2, 3, 5, 4, 5]
0 1 2 3
i = 4
arr=2k
item=5
i=2
arr=2k
item=5
i=1
S(2k, 5, 2)
2
arr=2k
item=5
i=0
S(2k, 5, 1)
main
(2k, 5, 0)

```
int[] arr = { 3, 5, 7, 8, 9 };
```

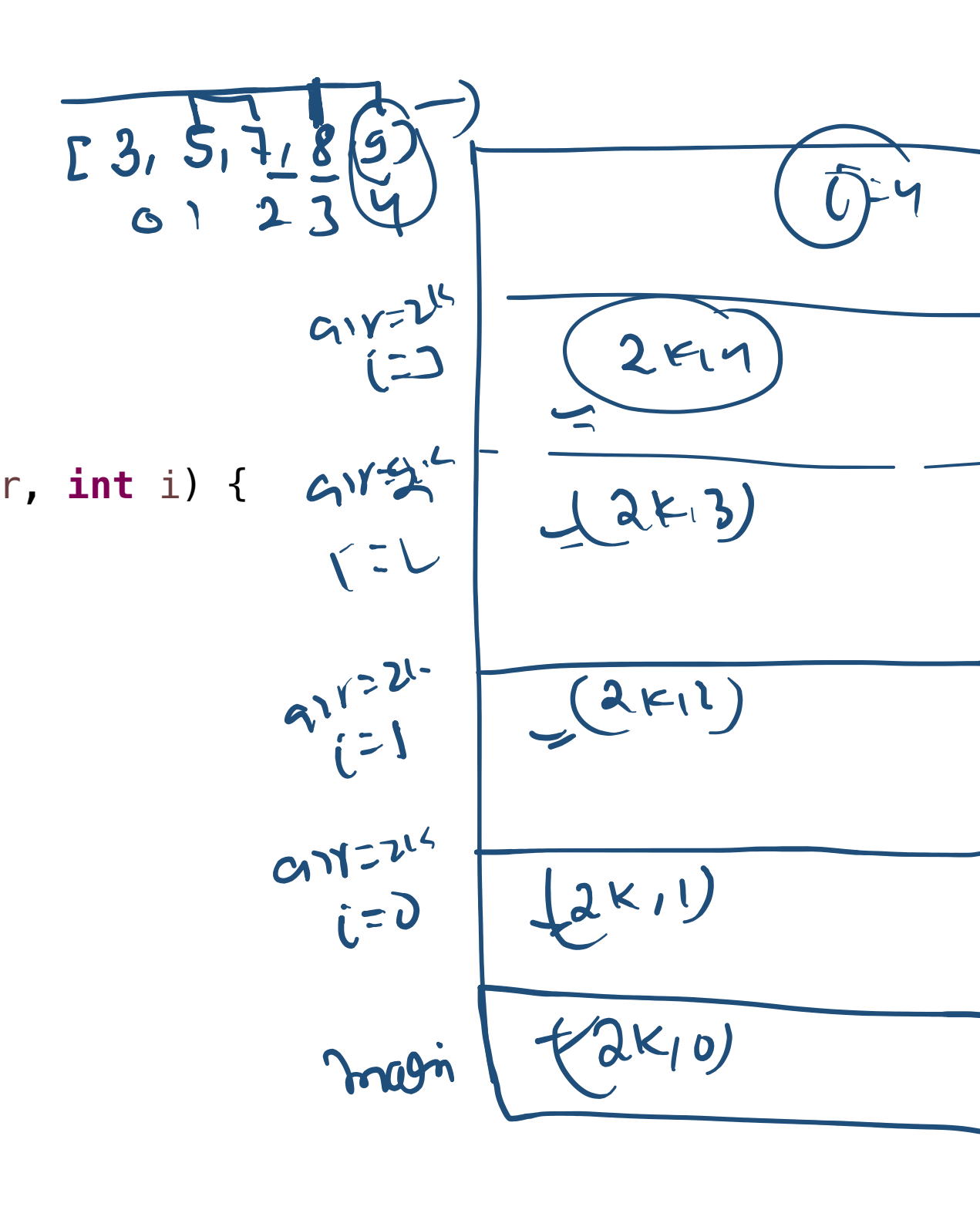
0 1 2 3 4

arr(i) > arr(i+1)

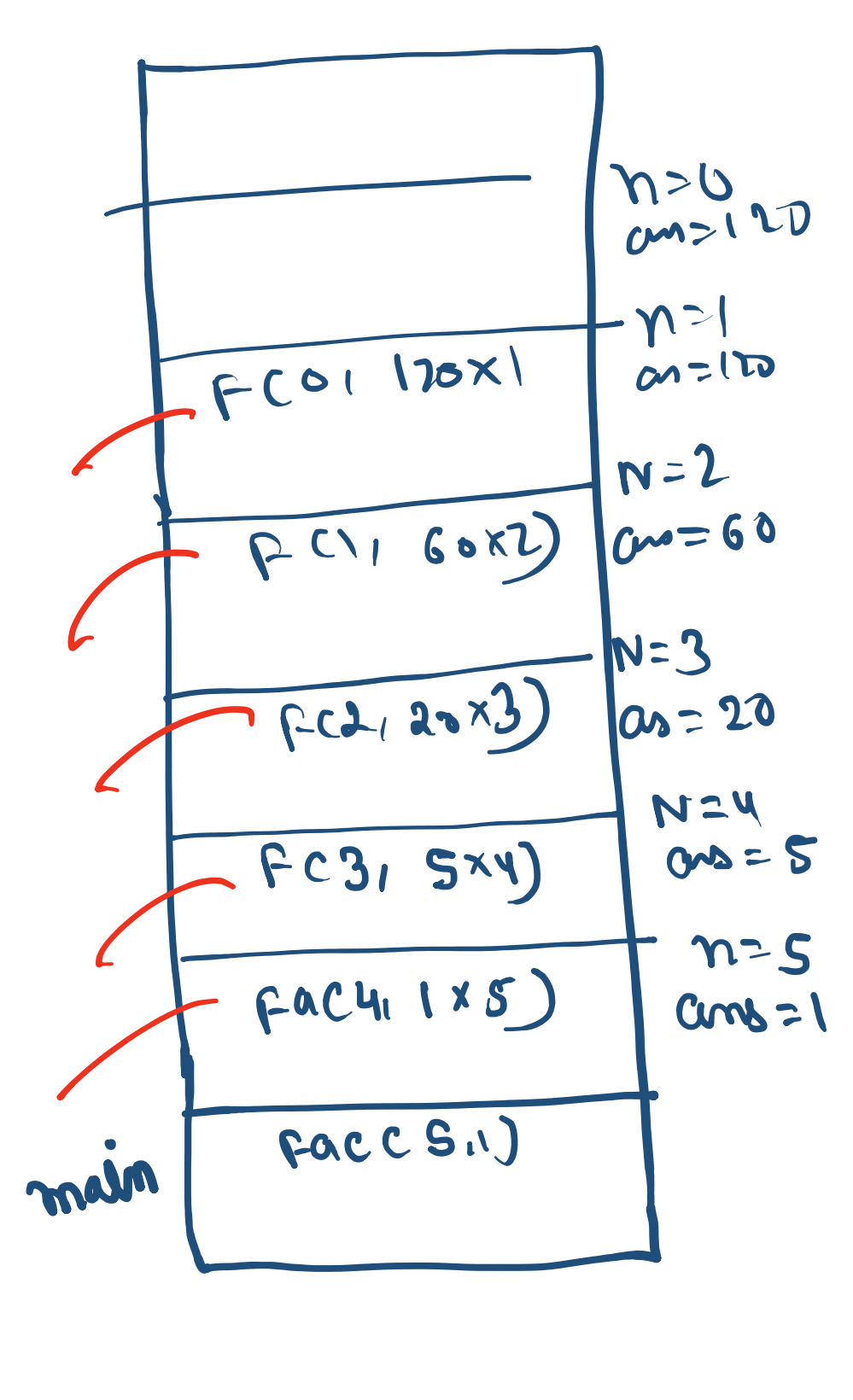
if (arr(i) > arr(i+1))
re false
}
sum(arr, i+1)

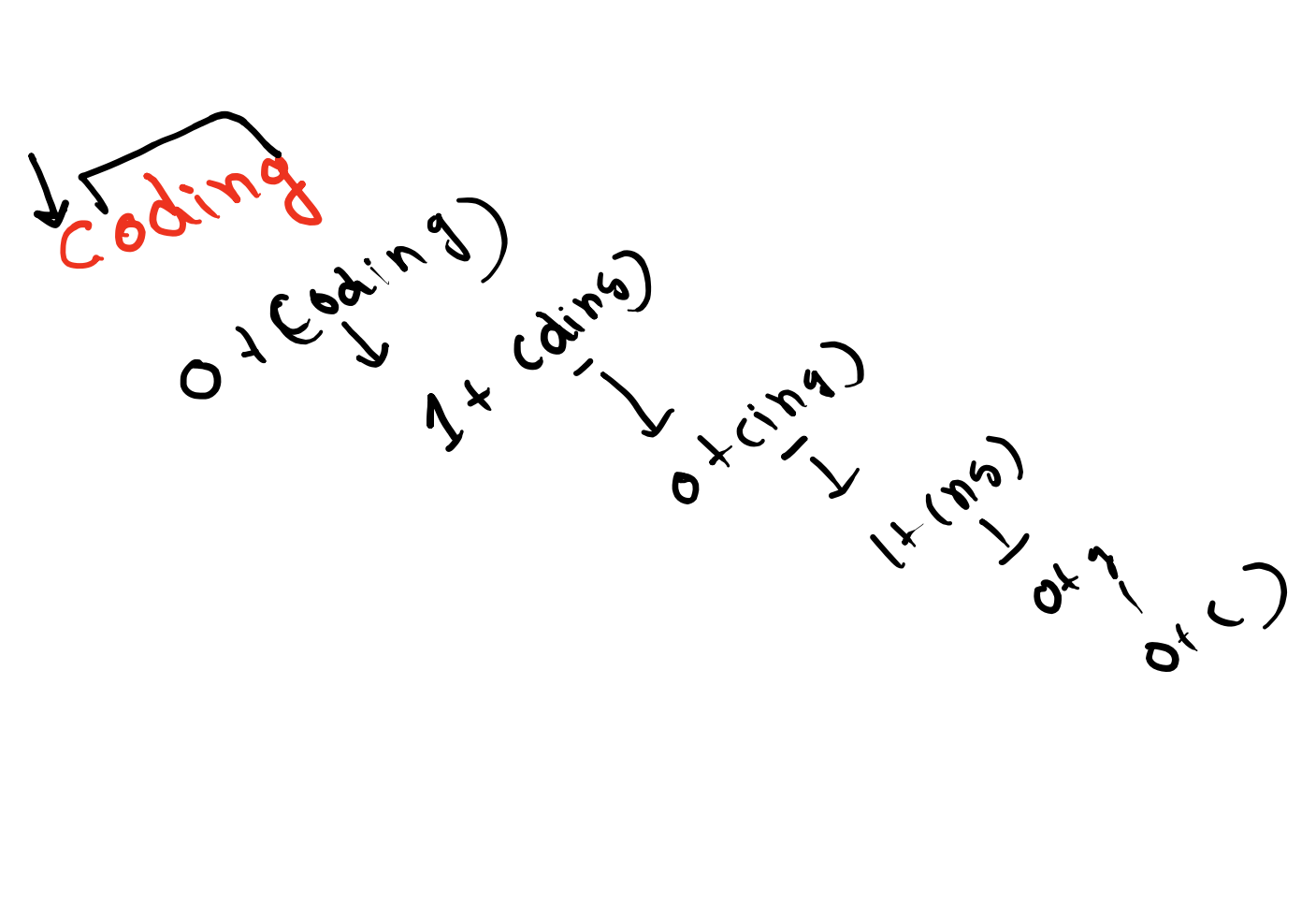
i=0 arr(0) < arr(1)
i=1 arr(1) < arr(2)
i=2 arr(2) < arr(3)
i=3 arr(3) < arr(4)
i=4

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        int[] arr = { 3, 5, 7, 8, 9 };
        System.out.println(Is_Sorted(arr,0));

    }

    public static boolean Is_Sorted(int[] arr, int i) {
        if (i == arr.length - 1) {
            return true;
        }

        if (arr[i] > arr[i + 1]) {
            return false;
        }
        return Is_Sorted(arr, i + 1);

    }
```

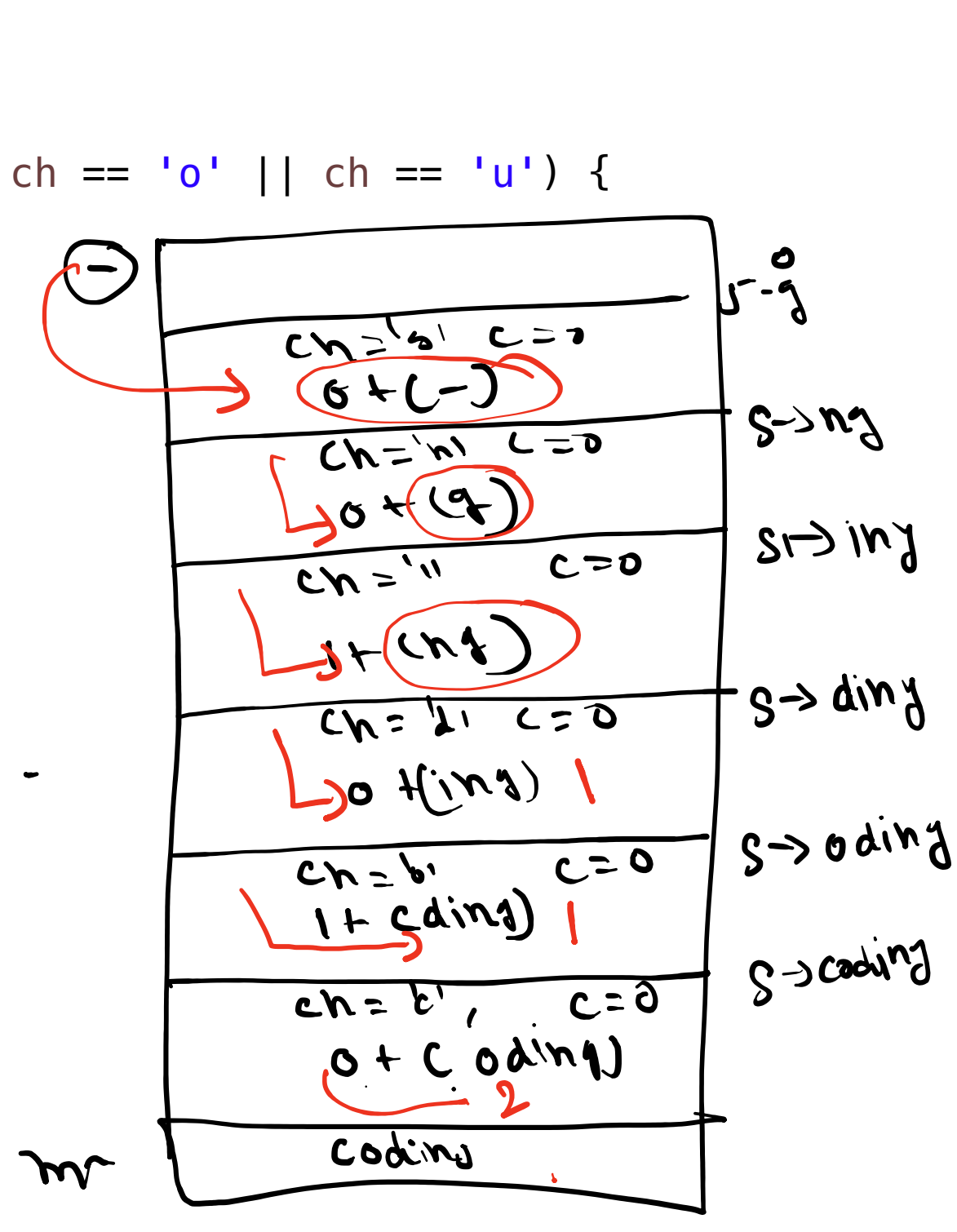

[3, 5, 7, 8, 9]
0 1 2 3 4
i=4
arr=2k
i=3
(2k, 4)
arr=2k
i=2
(2k, 3)
arr=2k
i=1
(2k, 2)
arr=2k
i=0
(2k, 1)
main
(2k, 0)

```
public static int fac(int n, int ans) {
        // base case
        if (n == 0) {
            return ans;
        }

        return fac(n-1, ans*n);
    }
```


n=0
ans=120
n=1
ans=120
F(0, 120×1)
N=2
ans=60
F(1, 60×2)
N=3
ans=20
F(2, 20×3)
N=4
ans=5
F(3, 5×4)
n=5
ans=1
fac(4, 1×5)
main
fac(5, 1)


Sub(1)
coding
0 + (coding)
1 + (ding)
0 + (ing)
1 + (ng)
0 + g
0 + ( )

```
public static int countVowels(String s) {

        char ch = s.charAt(0);
        int c = 0;
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {
            c = 1;
        }
        return c+countVowels(s.substring(1));

    }
```


s→g
ch='g' c=0
0 + ( )
s→ng
ch='n' c=0
0 + (g)
s→ing
ch='i' c=0
1 + (ng)
s→ding
ch='d' c=0
0 + (ing) 1
s→oding
ch='o' c=0
1 + (ding) 1
s→coding
ch='c' c=0
0 + (oding)
2
m
coding